Comp-3150-1 Winter 2026
Midterm 2 (solution)
Examiner: Dr. C. I. Ezeife   Given: Thursday, March 5, 2026

Student Name:___________________________________

Student Number:_____________________________

INSTRUCTIONS (Please Read Carefully)
Examination Period is 1 hours 20 minutes
Answer all questions. Write your answers in the spaces provided in the question paper. This is closed book and closed notes test.
Total Marks =50. Total number of sections = 2
Please read questions carefully! Misinterpreting a question intentionally or unintentionally results in getting a "ZERO" for that question. Good Luck!!!

CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY
I confirm that I will keep the content of this assignment/examination confidential.
I confirm that I have not received any unauthorized assistance in preparing for or doing examination. I confirm knowing that a mark of 0 may be assigned for copied work.
For Online Test/Examination in Comp 3150 Fall 2020: (additional rules to be observed):
1. I confirm that I agree to write this final examination as a closed book examination.
2. I confirm that I am the student with the name and student id signed below.
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For marking purposes only (This part not to be filled by students)

Question	Mark
Section A (10 marks for 10 multiple choice questions)	/10
Section B	
Que 1 (10 marks)	/10
Que 2 (15 marks)	/15
Que 3 (15 marks)	/15
Total	/50

Section A

10 marks for 10 Multiple Choice Questions. Each question in this section is worth 1 mark.

Marking scheme: 1 mark for each correct multiple choice.

For Questions 1 – 10, circle the most correct answer for each question.

Consider a relation R(V, W, X, Y, Z) where the following functional dependencies exist:
(V, W, X) -> {Y, Z}
Y -> {X}

1. **The relation R(V, W, X, Y, Z) *above is in 2NF:***
   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

2. **The relation R(V, W, X, Y, Z) *above is not in 3NF:***
   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

3. **The relation R(V, W, X, Y, Z) *above is not in BCNF:***
   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

Given a relation *R1(V, W, Z, F)* with the following functional dependencies:
(V, W, Z) -> {F}
F -> W

4. **Relation *R1(V, W, Z, F)* is in 3NF:**

   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

5. **The relation *R1(V, W, Z, F) above is not in BCNF:***

   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

Given the relation Course(C1410, C2140, C3150, M1020) with the following functional dependencies:
(*C1410, C2140*)* -> *{* C3150, M1020*}
C2140 -> *{* M1020*}*

6. **The relation Course(C1410, C2140, C3150, M1020) *above is not in 3NF:***

   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

7. **The relation Course(C1410, C2140, C3150, M1020) *above is in 2NF:***

   a. **TRUE**
   b. **FALSE**
   c. **MAY BE**
   d. **ALL OF THE ABOVE**
   e. **NONE OF THE ABOVE**

JOB

Jname	Jobid	location	IdD
FixX	1	London	4
FixY	2	Windsor	4
FixZ	3	Toronto	4
IT	10	Ottawa	5
Manage	20	Chatham	1
HumanResource	30	Ottawa	1

8. **Which of the following statements is correct about the table JOB?**

   a. **It does not contain update anomaly**

   b. **It does not contain insert anomaly**

   c. **It does not contain delete anomaly**

   d. **All of the above**
   e. **None of the above**

9. **Which of the following create table command is given most correctly for creating the table JOB with a foreign key attribute IdD that refers to another table DEPARTMENT's attribute idD that is of numeric type?**

   a. **CREATE TABLE JOB(Jname VARCHAR2(15),**
   Jobid CHAR(1), location VARCHAR2(15), IdD NUMBER(3));

   b. **CREATE TABLE JOB(Jname VARCHAR2(15),**
   Jobid DATE, location VARCHAR2(15), IdD NUMBER(3),
   PRIMARY KEY (Jobid), FOREIGN KEY (IdD) REFERENCES
   DEPARTMENT(IdD));

   c. **CREATE TABLE JOB(Jname VARCHAR2(15),**
   Jobid NUMBER NOT NULL, location VARCHAR2(15),
   IdD VARCHAR2(3), PRIMARY KEY (Jobid), FOREIGN KEY (IdD)
   REFERENCES DEPARTMENT(IdD));

   d. **All of the above**
   e. **None of the above**

10. **The effect of the following SQL instruction on the JOB table, when the only known IdD instances in the DEPARTMENT table are 1, 4 and 5 is ________ .**

    INSERT INTO JOB VALUES('Finance', 4, 'London', 3);

    a. **Insert an additional tuple in the JOB table with values specified.**
    b. **Update the second tuple of the JOB table to have the values specified.**
    c. **Raise an exception to indicate that there is a primary key constraint violation.**
    d. **Raise an exception to indicate that there is a referential integrity violation.**
    e. **None of the above**

Section B (40 marks):
This section has 3 questions. Answer all questions in this section.

1. **(10 marks)** Given the following relation schema T, with attribute A as the primary key, answer the questions below.

T Relation

A	B	C	D
a1	b2	c1	d1
a2	b2	c2	d2
a3	b2	c1	d1
a4	b2	c3	d1

(i) Write all the functional dependencies FDs you see exist between the attributes of this relation above?

(ii) What update, (iii) delete and (iv) insertion anomalies may or not be present in this T relation above. Explain with specific examples using your listed FDs, keys and the specific attributes affected in this database.

Que 1 (2.5 marks for each of i to iv for a total of 10 marks)

(i) FDs (2.5 marks)	FD1: A -> {B, C, D} FD2: D -> B; FD3: C -> D; FD4: C → B; Since for FD2, every tuple with D value d1, its B value is b2 and every tuple with D value d2, its B value is b2. Also, for FD3, for every tuple with C value of c1, it has a D value of d1. This indicates that the primary key does not fully functionally determine all attributes and thus, this table is not in 3NF and contains anomalies. Detailed marking scheme: Duck -0.5 for each of the missing FDs. Duck -2.5 if none is correct.
(ii) Update anomalies (2.5 marks)	Since D -> B, if there is need to change the D value (e.g. d1 has b2 value for its B attribute and we want to change its D value to d3, we must change all other tuples with D value d1 to d3 and have all their B values remain b2), thus several tuples having the D value d1 in the table will have to be updated or there is a violation of the functional dependency (D -> B) in the database that must exist. This is update anomaly. Detailed marking scheme: Duck -2.5 for poor explanation of either redundant multiple update or violation of FDs. Duck -1 if a decent explanation is provided.

(iii) Delete anomalies (2.5 marks)	In the table above, if we delete the 1 tuple with D value of d2, we no longer know what B values are associated with a D value of d2. This is delete anomaly. Detailed marking scheme: Duck -2.5 for poor explanation of losing the relationship between attributes of FDs should the last one be deleted. Duck -1 if a decent explanation is provided.
(iv) Insertion anomalies (2.5 marks)	If a new tuple is inserted whose D and B values are not yet known, we might not be able to insert this tuple until we can assign them a D value with matching B value that maintains the FD (D -> B). This is insert anomaly. Thus, we cannot insert a NULL value for D although it is not a key. Detailed marking scheme: Duck -2.5 for poor explanation of not being able to insert NULL value for B or insert the tuple until a B value is known due to the FD B -> A. Duck -1 if a decent explanation is provided.

2. **(15 marks)** Given the following four relations for a simple restaurant management record database application where cost of meals ordered by customers are tracked for each food order. Total number of orders received by each customer is known from the customer record at all times.

Customer(cid, cname, address, numorder)
Chef(chid, chname, specialization, address)
Meal(mealid, meal_name, cost)
Serves(chid, cid, mealid, mdate)

Here, the primary key for each relation is underlined. The attributes cid stands for customer id, chid is chef id, mealid is meal id, numorder is the cumulative number of meal order the customer has made to this restaurant managed by a chef (e.g., 50). All other attributes are self explanatory.
A small database state of this database is:

CUSTOMER

cid	cname	address	Numorder
1111	John Smith	Windsor	100
2222	Mary Pert	Windsor	50

CHEF

chid	chname	specialization	Address
1	Mana Patel	celebrity	Windsor
2	Pat Goodman	fish	London

MEAL

mealid	meal_name	Cost
23567	Chicken_rice	20.45
55555	Fish_potato	30.50

SERVES

chid	cid	mealid	Mdate
1	1111	23567	10-Jan-21
2	2222	55555	20-Aug-20

a. Write the SQL DDL instructions to create all 4 tables of the above database in the Oracle DBMS on our delta.cs server specifying all constraints such as Key, entity and foreign key constraints as usual within each instruction (not separately). **(5 marks)**

b. Using only the data in the above instance (not a new instance formed by you), write the SQL instructions to load (insert) all the data above in the 4 tables you created. **(5 marks)**

c. Write the SQL query for the following query posed on this database, also write the results of the query and the tables involved in answering the query. **(5 marks)**

Print all meals names and date they are ordered (meal_name, mdate) for customer named 'Mary Pert'.

(5 marks)

Que 2a (5 marks): Creating Tables

```
CREATE TABLE CUSTOMER
(CID   VARCHAR2(10) NOT NULL,
Cname VARCHAR2(15),
Address VARCHAR2(15),
Numorder NUMBER(8),
PRIMARY KEY(Cid));

CREATE TABLE CHEF
 (CHID VARCHAR2(10) NOT NULL,
Chname VARCHAR2(15),
Specialization  VARCHAR2(15),
Address  VARCHAR2(15),
PRIMARY KEY(CHID));

CREATE TABLE MEAL
(MEALID VARCHAR2(10) NOT NULL,
Meal_name  VARCHAR2(15),
Cost NUMBER(8, 2),
PRIMARY KEY(MEALID));
```

```
CREATE TABLE  SERVES
(CHID   VARCHAR2(10) NOT NULL,
CID  VARCHAR2(10) NOT NULL,
MEALID VARCHAR2(10) NOT NULL,
PDate DATE,
PRIMARY KEY(Chid, Cid, MEALID),
FOREIGN KEY (Chid) REFERENCES CHEF(Chid),
FOREIGN KEY (Cid) REFERENCES CUSTOMER(Cid),
FOREIGN KEY (MEALID) REFERENCES MEAL(MEALID));

COMMIT;
```

Detailed marking scheme: It is -1 for each of the 4 tables not created well or -0.5 for any incorrect constraint or attribute up to two for each table.

Que 2b (5 marks): Inserting Data into the Database Tables

Solution (b)

```
INSERT INTO CUSTOMER
VALUES ('1111', ‘John Smith’, 'Windsor', 100);
COMMIT;

INSERT INTO CUSTOMER
VALUES ('2222', ‘Mary Pert’, 'Windsor', 50);
COMMIT;

INSERT INTO CHEF
VALUES (‘1’, ‘Mana Patel’, ‘Celebrity, `Windsor’);
COMMIT;

INSERT INTO CHEF
VALUES (‘2’, ‘Pat Goodman, ‘Fish’, `London’);
COMMIT;

INSERT INTO MEAL
VALUES ('23567', ‘Chicken-rice’, 20.45);
COMMIT;

INSERT INTO MEAL
VALUES ('55555', ‘Fish_potato’, 30.50);
COMMIT;

INSERT INTO SERVES
VALUES (`1’, ‘1111’, '23567', ‘10-Jan-21’);
COMMIT;
```

```
 INSERT INTO CHEF
VALUES ('2', '2222', '55555', '20-Aug-20');
COMMIT;
```

Detailed marking scheme: It is -1 for each of the 4 tables not having all its data loaded well or -0.5 for any line of data with error or missing up to 2 per table.

Question	Query	Tables needed to answer query and Result
2c (5 marks)	Detailed marking scheme: It is 3marks for query, list of tables needed is 1 mark, and result of query is 1 mark. Also, for query (spj) select conditions 2 (even if only one condition missing including important join conditions), project (1 mark). Print all meal names and date they are ordered (meal_name, mdate) for customer named 'Mary Pert'. SQL Query is: Select Meal.mname, Serves.mdate From Customer, Meal, Serves Where Customer.cid = Serves.cid And Meal.mealid = Serves.mealid And Customer.cname = 'Mary Pert;	Tables needed are: Customer, Meal, Serves Result of Query is: mname mdate Fish_potato 20-Aug-20

3. (15 marks)

Given the following WorkPlace relation schema, with Ssn as the primary key, answer the questions below.

WorkPlace

Ssn	name	locker#	rank	wage	hours
1	Bob	28	2	20	40
2	Bill	21	2	20	30
3	Pat	15	1	10	30
4	Tony	15	1	10	32
5	Namdi	15	2	20	40

a. Is this table in 3NF? Explain your first answer to a) using all applicable functional dependencies (FD); If table not in 3NF, normalize it into 3NF and show all the tables in your new decomposed database with all FDs in the new decomposed database.

(**6 marks**) [split as 1; 2; 1.5, 1.5]

b. Provide one valid tuple instance that can be inserted into this original database. (**3 marks**)

c. Provide one SQL query for updating a record in the original table. (**3 marks**)

d. Provide one SQL query for (i) deleting all tuples in the original table, then (ii) provide one SQL query for deleting the original table schema from your database. (**3 marks**)

<table>
<tr><td>Question</td><td>Answers</td></tr>
<tr><td>a.
(6 marks)</td><td>Discussion
WorkPlace(Ssn, name, locker#, rank, ~~wage~~, hours) (not normalized because of violating FD: rank → wage)
How to normalize is usually, to decompose to have the right hand attributes of the violating FDs removed from the main table, while re-inserting the violating FD as a table with the left attribute as the primary key.</td></tr>
<tr><td>Is this table in 3NF?, (1)</td><td>NO, it is not n 3NF.</td></tr>
<tr><td>Explain your answer above with FDs (2)</td><td>Violating FD is: rank -> {wage}
This is because for every tuple with rank value 1, its wage is 10 and every tuple with rank value 2, its wage is 20. This indicates that the primary key does not determine all non-key attributes non-transitively, and thus, this table is not in 3NF and may contain anomalies.</td></tr>
<tr><td>normalize into 3NF showing all tables (1.5)</td><td>To decompose into 3NF relations, we need to remove the violating FD2 from the original table by deleting the attribute wage from the original relation and we create a new relation with the FD2 to have the following two normalized relations.
WorkPlace1(Ssn, name, locker#, rank, hours)
WorkPlace2 (rank, wage)</td></tr>
<tr><td>Show FDs In new normalized relations (1.5)</td><td>It can be seen that FD2 with refined FD1 are preserved by this decomposition in this database since the FDs in the decomposed database are:
FD1: Ssn -> {name, locker#, rank, hours}
FD2: rank -> {wage}

Detailed marking scheme: It is -1 for not knowing table is not in 3NF. It is -2 marks for not discussing well with FDs. It is -3 marks for not correctly decomposing to normal form based on violating FDs, and showing the new FDs in the normalized relations (-1.5 for normalized relations and -1.5 for the FDs of normalized relations). Lose 1 in the decomposition mark of 1.5 partially correct decomposed relation.</td></tr>
</table>

b. one valid tuple instance (3 marks)	Any tuple that is inserted has to conform to all the FDs in the database which are FD1 and FD2. Thus, a valid tuple that can be inserted into the original WorkPlace table is: 6 Mimi 17 1 10 50 Detailed marking scheme: It is -3 mark for any insertion causing an anomaly.
c. SQL query to update a record (3 marks)	UPDATE WorkPlace SET hours = 25 WHERE name = 'Bob'; Detailed marking scheme: It is -1 mark for missing each of the 3 parts of Tablename, SET attrs, and where condition.
d. (3 marks) i.SQL query to delete all records from the original table ii.SQL query to remove or delete the original table.	 DELETE FROM WorkPlace; DROP TABLE WorkPlace CASCADE; Detailed marking scheme: It is -1.5 mark for missing each of the Delete or Drop table instructions. Note that Drop table can be with or without cascade and may also be in the form of cascade constraints accepted by our system.